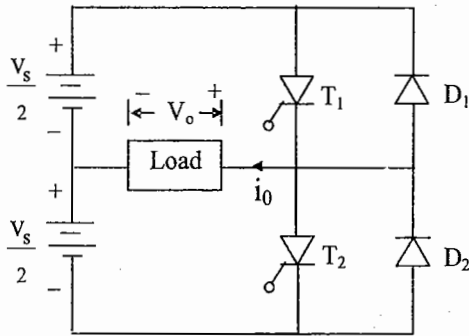


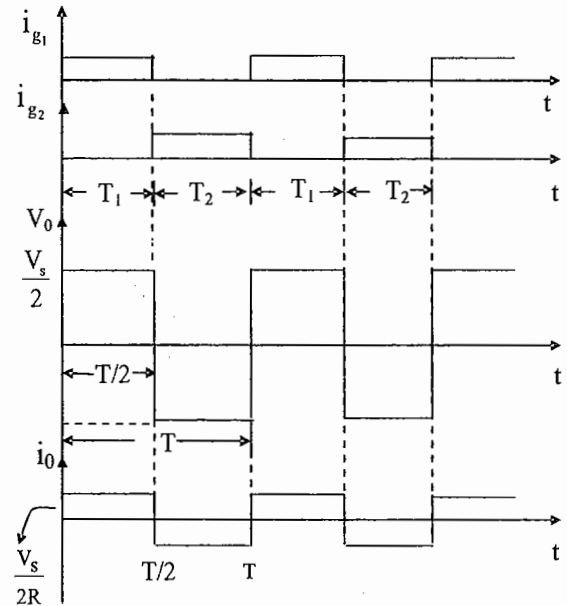
- It is a static power electronic circuit which converts DC to variable AC i.e., Variation in Magnitude of voltage, frequency and number of phases.
- Phase controlled rectifiers when operated with $\alpha > 90^\circ$, they are line commutated inverters. It transfers the energy from DC to an existing AC supply Network. The output AC voltage, frequency and no of phases cannot be controlled.

1 - ϕ Half Bridge Inverter:



- Both thyristors are forward biased. So, triggering is given to them so that only one conducts at a time.
- It consists of two SCR's, two diodes and three-wire supply. For $0 < t \leq \frac{T}{2}$, thyristor T_1 conducts and the load is subjected to a voltage $\frac{V_s}{2}$ due to the upper voltage source $\frac{V_s}{2}$. At $t = \frac{T}{2}$, thyristor T_1 is commutated and T_2 is gated on. During the period $\frac{T}{2} < t \leq T$, thyristor T_2 conducts and the load is subjected to a voltage $\left(-\frac{V_s}{2}\right)$ due to the lower voltage source $\frac{V_s}{2}$.
- It is seen from the output wave forms that the load voltage is an alternating voltage waveform of amplitude $(V_s/2)$ and frequency

$\frac{1}{T}$ Hz. The frequency of the inverter output voltage can be changed by controlling 'T'.



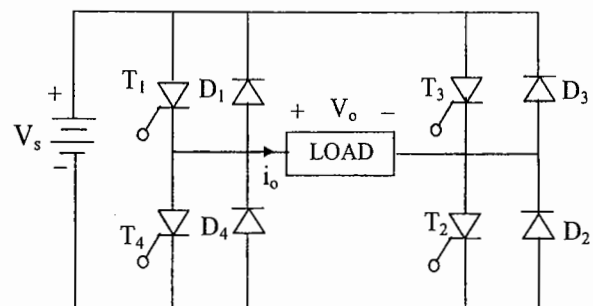
- Inverter operating principle based on Forced Commutation. The output wave forms are square wave forms. They will be passed through filters to eliminate unwanted harmonics.
- Anti-parallel diodes are required for all the loads except resistive load.

$$V_{or} = \frac{V_s}{2}$$

Disadvantage:

- At any time output voltage is half of the available supply voltage. So, the source utilization factor will be 50%.

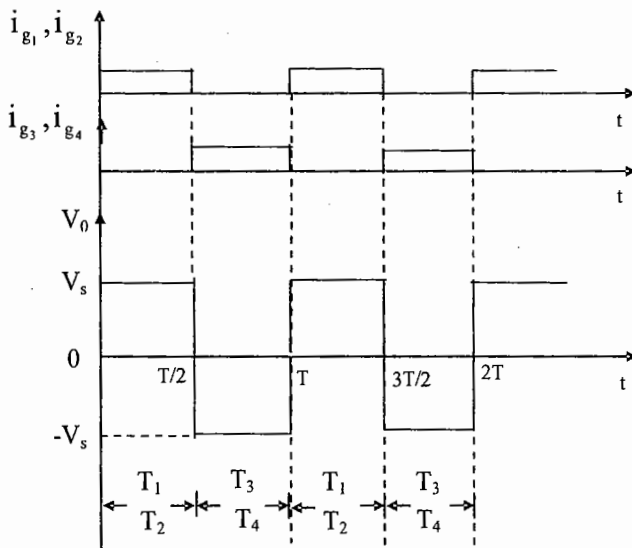
1 - ϕ Full Bridge Inverter:



- If all the thyristors conduct at a time then the source will be short circuited.
- Choose a suitable combination of two thyristors triggering at different times so that current always passes through load.

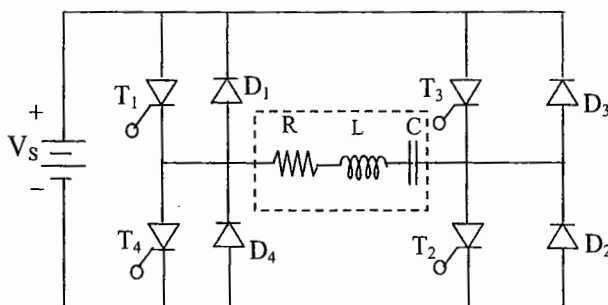
$$V_{or} = V_s$$

- For a full bridge inverter, when T_1, T_2 conducts, load voltage is V_s and when T_3, T_4 conduct load voltage is $-V_s$. frequency of the output voltage can be controlled by varying the periodic time 'T'.
- At any time o/p voltage is same as the supply voltage. So the source utilization factor will be 100%.
- For a given input dc supply, load or output voltage is two times, output power is four times in the Full bridge inverter compared to half bridge inverter.



- As the energy is fed back to the dc source when these diodes conduct, these are called feed back diodes.

Steady State Analysis of Single Phase Inverter:



- In under damped, loading condition ($X_c > X_L$) Load commutation will be taken place. No external circuit is required for commutation. The current through thyristors coming to zero, sub sequent conduction of anti-parallel diodes to apply reverse voltage across thyristor are due to nature of the load. So, it is a Load Commutation'.

- For reliable commutation of thyristors, the anti-parallel diodes conduction duration.

$$\left(\frac{T}{2} - t_1\right) = t_c > t_q$$

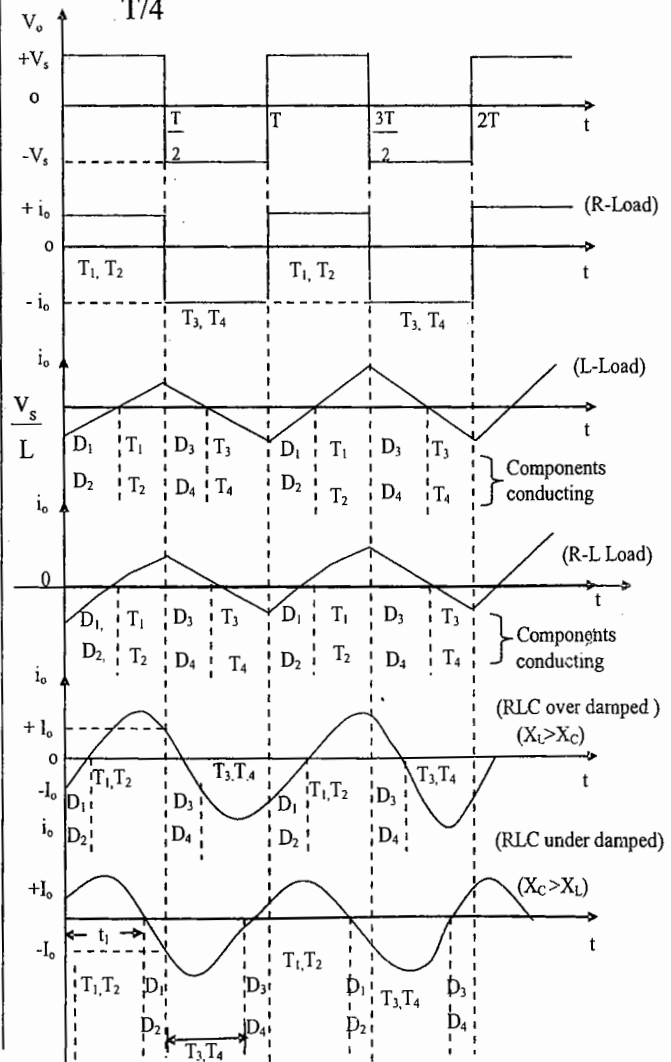
- If inverter is feeding a pure inductive load, the following are valid:

(1) Nature of the current = Triangular.

(2) Slope of current variation $I_p = \left(\frac{V_s}{L}\right) \left(\frac{T}{4}\right)$

(3) $I_{p-p} = 2 I_p$.

(4) Conduction time of either diode/thyristor = $T/4$



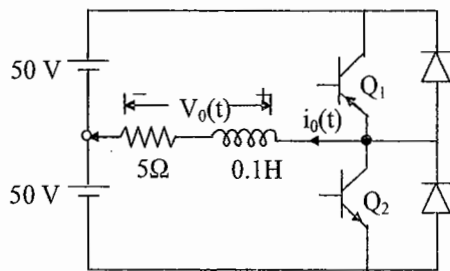
Workedout Examples

01. Which of the following statement/statements is /are correct in connection with inverters:

- (a) VSI and CSI both require feedback diodes.
- (b) Only CSI requires feedback diodes.
- (c) GTO's can be used in CSI
- (d) Only VSI requires feedback diodes.

Common Data for linked Questions 02 & 03

The inverter shown in figure is operated at a frequency of 50 Hz. Q_1 is ON for the 1st half cycle and Q_2 for the next, and so on. All devices are assumed ideal.



02. $i_o(t)$ at the end of the 1st half cycle (in amp) is
 (a) 10 (b) 3.935
 (c) 5 (d) 1.572

Sol: At $t = 0^-$ assume that $i_o(t) = 0$. Let Q_1 be on at $t = 0$. For $t \geq 0^+$, $Ri_o + L \frac{di_o}{dt} = E$ ($R = 5\Omega$, $L = 0.1H$, $E = 50V$)

$$i_o(t) = \frac{E}{R} \left(1 - e^{-\frac{Rt}{L}} \right) = 10 (1 - e^{-50t}) \text{ A.}$$

For 50Hz; the 1st half cycle ends at $t = 10 \text{ ms.}$ $i_o(t)$ at the end of the 1st half cycle
 $= 10 (1 - e^{-0.5}) = 3.935 \text{ A}$

Fourier analysis of 1-φ Inverter Output Voltage:

1-φ Half bridge inverter

$$V_o = \sum_{n=1,3,5,\dots}^{\infty} \left(\frac{2V_s}{n\pi} \right) \sin n\omega t$$

$$i_o = \sum_{n=1,3,5,\dots}^{\infty} \left(\frac{2V_s}{n\pi Z_n} \right) \sin (n\omega t - \phi_n)$$

1-φ Full bridge inverter

$$V_o = \sum_{n=1,3,5,\dots}^{\infty} \left(\frac{4V_s}{n\pi} \right) \sin n\omega t$$

$$i_o = \sum_{n=1,3,5,\dots}^{\infty} \left(\frac{4V_s}{n\pi Z_n} \right) \sin (n\omega t - \phi_n)$$

- All the above formulae are in $V_m \sin \omega t$ form (V_m) = Maximum value of voltage.

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$Z_n = \sqrt{R^2 + \left(n\omega L - \frac{1}{n\omega C} \right)^2}$$

Z_n = load impedance at frequency $n.f$

ϕ_n = phase angle.

$$\phi_n = \tan^{-1} \left(\frac{n\omega L - \frac{1}{n\omega C}}{R} \right) \text{ rad}$$

- If the order of harmonics is increased, then the impedance is increased.

(1) Harmonic factor of nth harmonic ($H F_n$):-

- It is defined as the ratio of rms value of n^{th} harmonic voltage component, to the rms value of the fundamental voltage component.

$$(H.F)_n = \left| \frac{V_n}{V_{01}} \right|$$

Where V_n = rms value of the n^{th} harmonic component of output voltage.

V_{01} = rms value of fundamental component of output voltage.



(2) Total Harmonic Distortion (THD):

- It is defined as the ratio of rms value of all the harmonic components, to the rms value of fundamental component.

$$\text{THD} = \frac{V_{oh}}{V_{o1}}$$

where $V_{oh} = \sqrt{V_{or}^2 - V_{o1}^2}$

V_{oh} = rms value of all harmonic components present in the inverter output voltage.

V_{or} = rms value of inverter output voltage, including fundamental plus all the harmonics.

$$\text{THD} = \frac{\sqrt{V_{or}^2 - V_{o1}^2}}{V_{o1}} = \left[\left(\frac{V_{or}}{V_{o1}} \right)^2 - 1 \right]^{1/2}$$

- THD is a measure of the waveform distortion. Lower the value of THD, lower is the amount of distortion in voltage (or current) waveform.

$$\text{THD} = \frac{\sqrt{\sum_{n=2,3,\dots}^{\infty} V_n^2}}{V_{o1}}$$

(3) Distortion factor:

It is defined as the ratio of fundamental rms output voltage to the total rms output voltage.

$$\text{D.F.} = \frac{V_{o1}}{V_{or}}$$

Distortion factor of n^{th} harmonic.

$$(\text{D.F.})_n = \frac{V_n}{n^2 V_{o1}}$$

Workedout Examples

03. The output voltage of a 3-phase voltage source inverter contains 5th and 7th harmonics. Assume the output is balanced.

If $V_a = V_{1m} \sin(\omega t) + V_{5m} \sin(5\omega t) + V_{7m} \sin(7\omega t)$ then V_b can be expressed as

(a) $V_b = V_{1m} \sin\left(\omega t - \frac{2\pi}{3}\right) + V_{5m} \sin(5\omega t) +$

$$V_{7m} \sin(7\omega t)$$

(b) $V_b = V_{1m} \sin\left(\omega t - \frac{2\pi}{3}\right) + V_{5m} \sin\left(5\omega t + \frac{2\pi}{3}\right) +$

$$V_{7m} \sin\left(7\omega t - \frac{2\pi}{3}\right)$$

(c) $V_b = V_{1m} \sin\left(\omega t - \frac{2\pi}{3}\right) + V_{5m} \sin\left(5\omega t - \frac{2\pi}{3}\right) +$

$$V_{7m} \sin\left(7\omega t + \frac{2\pi}{3}\right)$$

(d) None of these

04. A single phase full bridge VSI has a source voltage of 200V DC. The load consists of RLC in series where $R = 1\Omega$, $\omega L = 6\Omega$ and $\frac{1}{\omega C} = 7\Omega$. Identify the fundamental component of the load current from the following.

(a) $180 \sin(\omega t + 45^\circ)$

(b) $180 \sin(\omega t - 45^\circ)$

(c) $127.3 \sin(\omega t - 45^\circ)$

(d) $127.3 \sin(\omega t + 45^\circ)$

Sol: $V_{o1} = 0.9 \times 200 = 180$

$$I_{o1} = \frac{V_{o1}}{Z_1} = \frac{180}{\sqrt{1^2 + (6-7)^2}} = \frac{180}{\sqrt{2}}$$

$$\tan \phi = \frac{X_C - X_L}{R} \Rightarrow \phi = 45^\circ \text{ lead}$$

$$\Rightarrow i_o = 180 \sin(\omega t + 45^\circ)$$

05. A single phase full bridge VSI is feeding power to a resistive load of 2Ω . The fundamental output voltage is found to be 230 V (rms). Find the RMS value of switch and diode currents at fundamental frequency

(a) $I_{sw1} = 81.33 \text{ A}$ $I_{D1} = 0 \text{ A}$

(b) $I_{sw1} = 54.818 \text{ A}$ $I_{D1} = 0 \text{ A}$

(c) $I_{sw1} = 54.818 \text{ A}$ $I_{D1} = 17.328 \text{ A}$

(d) None of these

Sol: Fundamental output current = $\frac{230}{2} = 115 \text{ A}$

$$I_{sw1, \text{rms}} = \frac{115\sqrt{2}}{2} = 81.33 \text{ A}$$

No diode action



06. A 3-phase full bridge VSI delivers power to a resistive load from a 450 V DC source. For a star connected balance load of $10\Omega/\text{ph}$. Find the RMS value of the load current under 180° conduction mode

- (a) 18.708 A (b) 13.23 A
(c) 18.371 A (d) 12.99 A

Sol:
$$I_{or} = \frac{1}{\pi} \left[\left(\frac{V_{dc}}{3R} \right)^2 \frac{\pi}{3} + \left(\frac{2V_{dc}}{3R} \right)^2 \frac{\pi}{3} + \left(\frac{V_{dc}}{3R} \right)^2 \frac{\pi}{3} \right]^{\frac{1}{2}}$$

$$= 18.708 \text{ A}$$

• Voltage Control in 1 - ϕ Inverter:

The output AC voltage of the Inverter can be controlled by any one of the following methods.

(a) External control of A.C. output voltage:

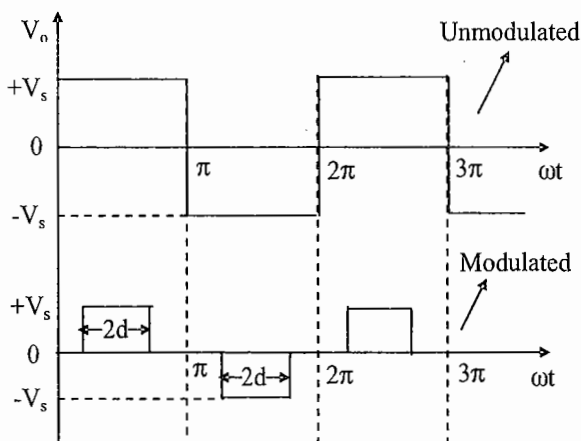
In this method an A.C. voltage controller is connected at the o/p of Inverter to get variable magnitude AC voltage.

(b) External control of D.C. Input voltage:

A chopper will be connected at the input of the inverter to control DC to variable DC voltage, this in turn controls the a.c. output voltage.

(c) Internal control of Inverters: pulse width Modulation (PWM) techniques are employed for the control of output voltage. In PWM method, width of the output pulse gets modulated while maintaining amplitude as constant.

Single pulse Modulation:



- The output waveform consists of only one pulse for half cycle. So it is a single pulse modulation.

- RMS value of output voltage,

$$V_{or} = \sqrt{\frac{1}{\pi} \int_{\frac{\pi}{2}-d}^{\frac{\pi}{2}+d} V_s^2 d\omega t}$$

$$V_{or}^2 = \frac{1}{\pi} (V_s)^2 \left(\frac{\pi}{2} + d - \frac{\pi}{2} + d \right)$$

$$V_{or}^2 = \frac{V_s^2 2d}{\pi}$$

$$V_{or} = V_s \sqrt{\frac{2d}{\pi}}$$

- Width of the output pulse can be controlled by varying the delay in firing angle and advance in commutation Angle.

$$V_o = \sum_{n=1,3,\dots}^{\infty} \left[\frac{4V_s}{n\pi} \sin\left(\frac{n\pi}{2}\right) \sin(nd) \right] \sin(n\omega t)$$

- RMS value of fundamental component,

$$V_{o1} = \left[\frac{\frac{4V_s}{\pi} \sin\left(\frac{\pi}{2}\right) \sin d}{\sqrt{2}} \right]$$

- By properly selecting the width of the pulse, one of the harmonic in output wave form can be eliminated.
- It is possible only when Amplitude of wave form becomes zero for that value of 'n'.

$$\sin nd = 0$$

$$nd = \pi \Rightarrow d = \frac{\pi}{n}$$

$$2d = \frac{2\pi}{n}$$

- If $2d = 0$; $V_{or} = 0$

$$2d = \pi; V_{or} = V_s$$

V_o varies from (0 to V_s)

- The o/p voltage wave form will have large amount of harmonics, particularly at low output voltage levels. Thus multi pulse modulation is preferred.

Workedout Examples

07. The output voltage of single phase FB VSI is controlled by using PWM with one pulse per half cycle. Determine the required approximate pulse width so that the fundamental RMS component is 70% of dc input voltage

Sol: $V_{o1} = \frac{2\sqrt{2}}{\pi} V_{dc} \sin(d) = \frac{70}{100} \times V_{dc}$

$$d = 51.05^\circ$$

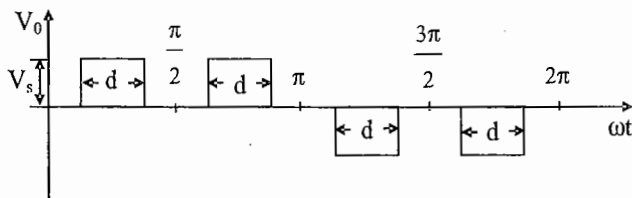
$$\text{Pulse width } 2d = 102^\circ$$

08. In single pulse modulation of PWM inverter, 3rd harmonic can be eliminated if pulse width is equal to

- (a) 30° (b) 60°
(c) 120° (d) 150°

Sol: Pulse width for eliminating nth harmonic $2d = 2\pi/n = 2\pi/3 = 120^\circ$

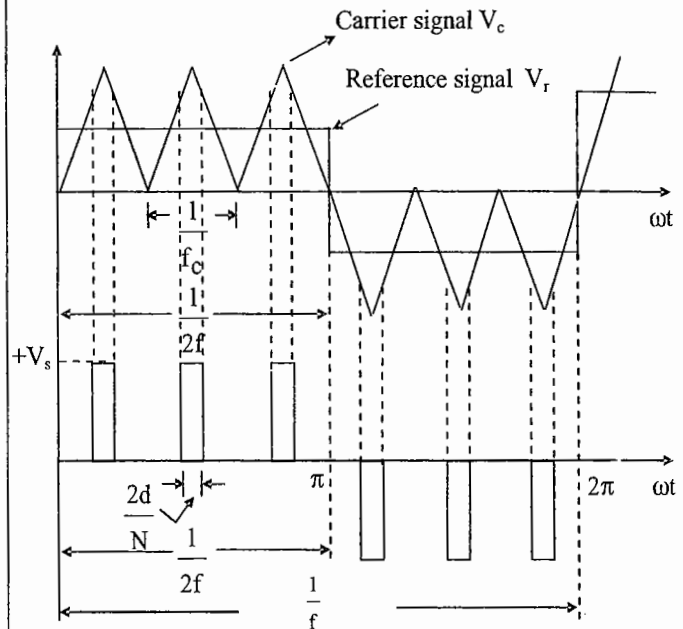
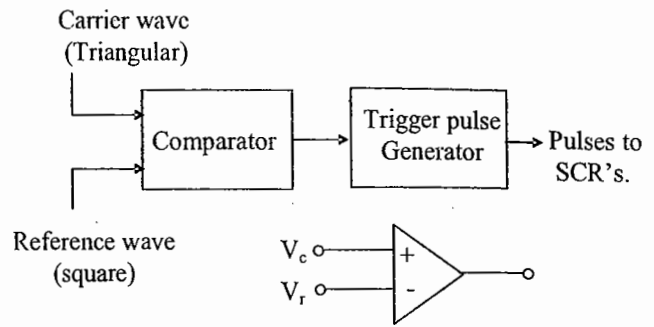
Simple Multiple Pulse Modulation:



- In multiple pulse modulation, all output pulses are identical having equal width of the pulses.
- RMS value of output voltage

$$V_{or} = V_s \sqrt{\frac{2d}{\pi}}$$

- If the number of pulses per half cycle increases then switching duty of thyristor increases. So, the no. of pulses would be economically selected to reduce the harmonics as well as minimizing the switching duty.
- The Amplitudes of Lower order harmonics are reduced but some of the higher order harmonics are significantly increased.



- Carrier wave – High frequency (Triangular wave).
- Reference wave – Low frequency (Square wave).
- Number of pulses per Half cycle.

$$N = \frac{\text{Length of half cycle of square reference wave}}{\text{Width of one cycle of triangular carrier wave.}}$$

$$N = \frac{1/2 f_r}{1/f_c} = \frac{f_c}{2 f_r} = \frac{\omega_c}{2 \omega} \Rightarrow N = \frac{f_c}{2 f_r}$$

- This means that if lower order harmonics are eliminated, the magnitude of higher order, harmonics would go up. But this is not a disadvantage, as higher order harmonics can be filtered out by the use of filters at the output terminals of the inverters.

Sinusoidal Pulse Width Modulation:

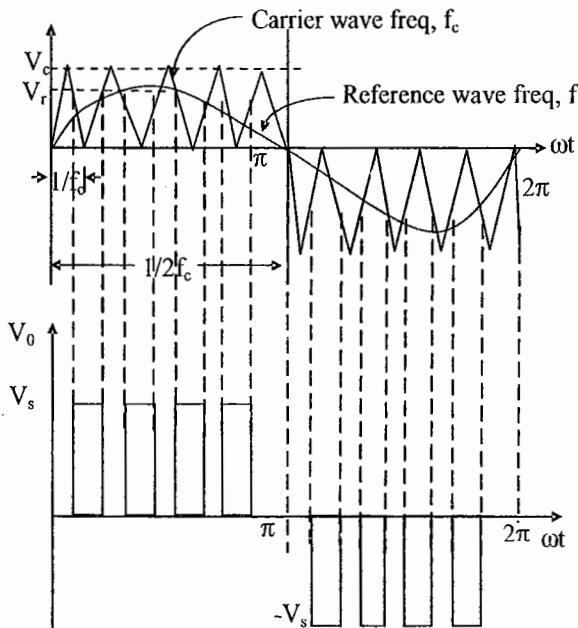
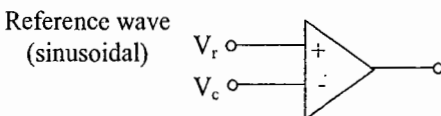
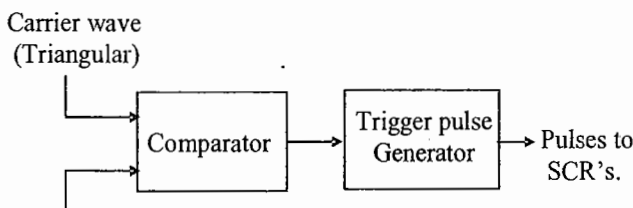


Fig. Output voltage waveforms with sinusoidal pulse modulation

- Carrier wave – High frequency (Triangular)
Reference wave – Low frequency (Sinusoidal)
- The output pulses are not identical i.e., they are having the variable width of the pulses.
- The width of the pulses varies in accordance with the magnitude of sinusoidal wave form.



- If peak of carrier is coincident with zero of the reference. number of pulses per Half cycle,

$$N = \frac{f_c}{2f_r}$$
- If zero of carrier is coincident with zero of reference, then number of pulses per Half cycle,

$$N = \frac{f_c}{2f_r} - 1$$

Modulation Index (M.I):

- The ratio of V_r/V_c is called the modulation index.
- If M.I. is less than 1, largest harmonic Amplitudes in output waveforms are associated of the order $2N \pm 1$.
- By Increasing the value of 'N', order of harmonic frequency will be increased which is easy to filter out.

Trapezoidal Pulse Width Modulation:

- Carrier wave – High frequency (Triangular)
- Reference wave – Low frequency (Trapezoidal)
- The width of the output pulses varies in accordance with the magnitude of trapezoidal wave form.

Triangular Pulse Width Modulation:

- Carrier wave – High frequency (Triangular).
Reference wave - Low frequency (Triangular).
- Width of the output pulses varies in accordance with the Magnitude of reference triangular waveform.

Workout Examples

09. The following statements relate to sinusoidal Pulse width modulated inverters.

- Pulse width is constant for all the pulses but amplitude varies.
- $\frac{V_r}{V_c}$ (modulation index), where V_r and V_c are the peak values of the sinusoidal reference voltage and triangular carrier voltage, should be equal to 1.
- When modulation index is less than 1, largest amplitudes occur for harmonics of order $2N \pm 1$ where N is the number of voltage pulses for a half cycle of the reference wave.

4. It is the high frequency harmonics in the output voltage which are more troublesome compared to the lower frequency harmonics.
5. When the triangular carrier wave has its peak coincident with Zero of the reference sinusoid, there are $\left(\frac{f_c}{2f}\right)$ voltage pulses per half cycle. (f_c and f are the Hz frequencies of the carrier and reference. Voltage pulses occur when the sinusoidal reference voltage is larger in magnitude than the carrier voltage).

True statements in the above are

- (a) 3, 4, 5 (b) 3, 5
(c) 1, 3, 5 (d) 1, 5

Sol: It is the amplitude which remains constant while pulse width varies. $\left(\frac{V_R}{V_C}\right)$ should be less than 1. Then harmonics of the order $(2N \pm 1)$ do have the highest amplitudes among all harmonics, but they can be filtered out relatively easily. Result (5) can be mathematically proved. Thus, statements (3) and (5) are true.

3- ϕ BRIDGE INVERTER:

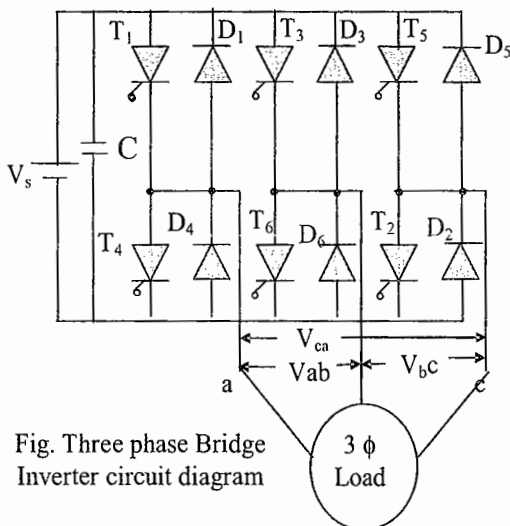


Fig. Three phase Bridge Inverter circuit diagram

180° Mode:

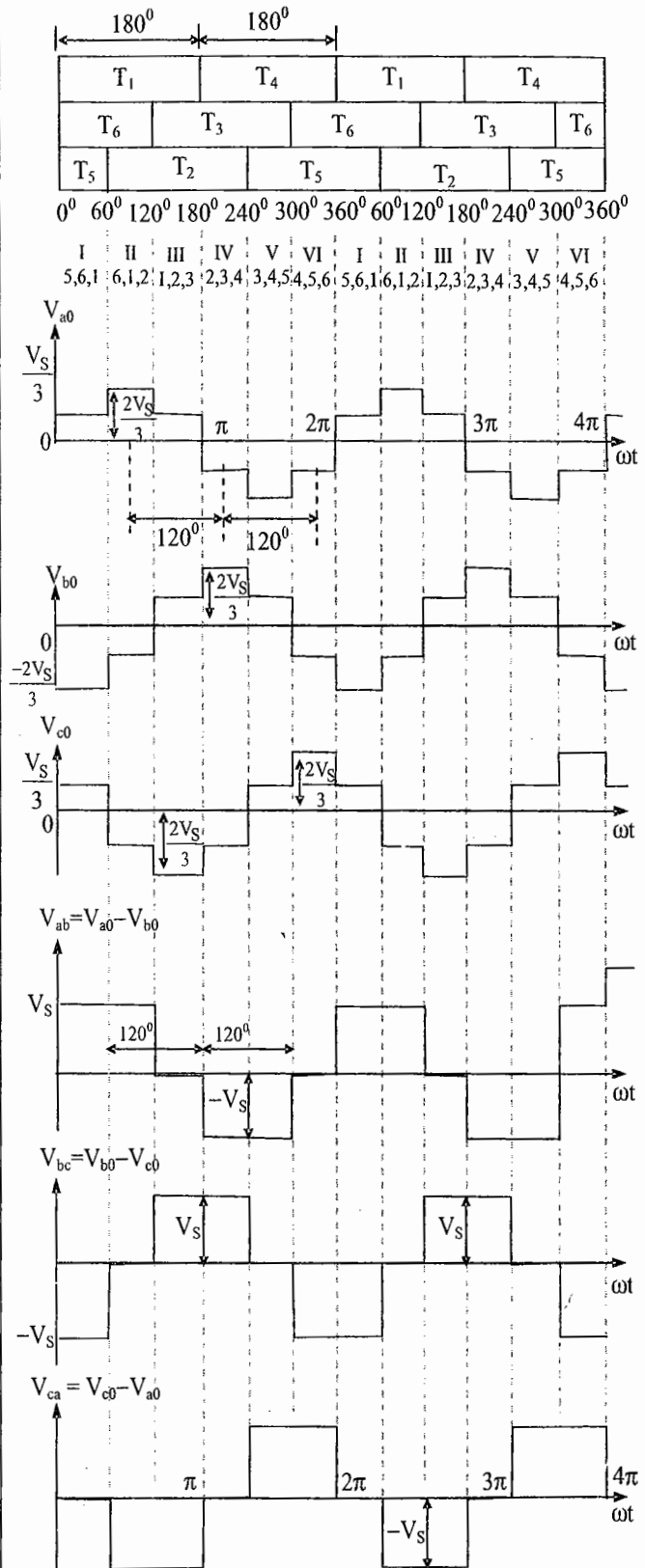
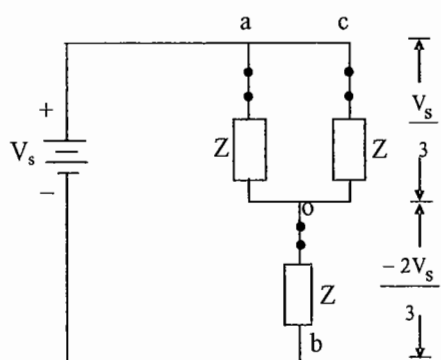


Fig. Voltage waveforms of phase voltages and line voltages for 180° mode

- Conduction time of each SCR = 180° .
- Phase angle between two sequential SCR's either from top group (or) from bottom group is 120° .
- At any time three SCR's would be there in the conduction.

$$V_{ab} = V_{ao} - V_{bo}$$

- Shape of phase voltage is three – stepped waveform.
- Shape of line voltage is Quasi – Square wave.
- In the I step (1,6,5) thyristors are turned ON.



- There is no time gap between commutation of out going SCR and conduction of incoming SCR. It leads to the simultaneous conduction of both incoming and outgoing SCR's, due to which source will be short circuited. Hence 120° mode of operation is used.

3 - ϕ 120° Mode VSI:

- Conduction time for each SCR = 120° .
- phase angle between two sequential SCR's either from top group (or) from bottom group is 120° .
- At any time, two SCR's will be there in conduction.
- Shape of phase voltage is Quasi square wave
- Shape of Line voltage is three – stepped waveform.
- At any time one of the SCR is neither connected to positive plate (nor) connected to negative plate. The particular phase voltage in that interval is uncertain if the load is unbalanced.

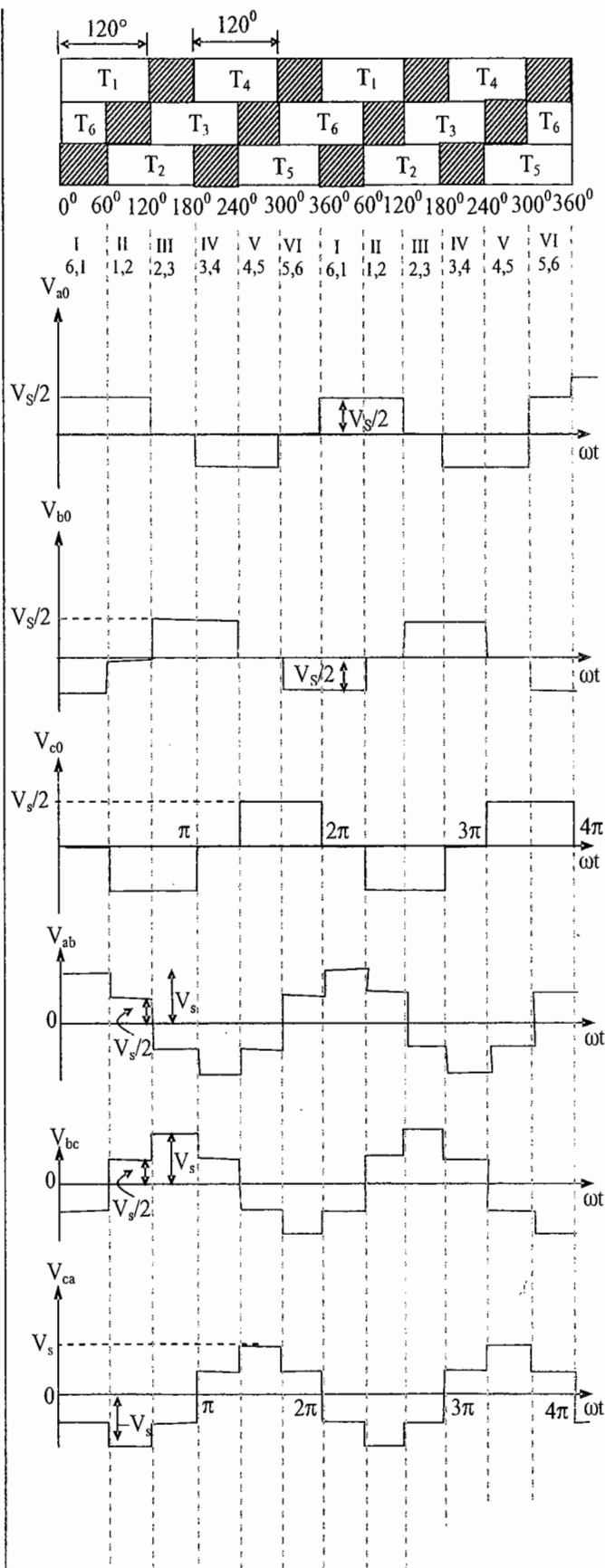


Fig. Voltage waveforms of phase voltages and line voltages for 120° mode

In 180° Mode VSI:-

$$V_{ab} = \sum_{n=1,3,\dots} \left\{ \frac{4V_s}{n\pi} \cos \frac{n\pi}{6} \right\} \sin n \left(\omega t + \frac{\pi}{6} \right)$$

$$V_{bc} = \sum_{n=1,3,\dots} \left\{ \frac{4V_s}{n\pi} \cos \frac{n\pi}{6} \right\} \sin n \left(\omega t - \frac{\pi}{6} \right)$$

$$V_{ca} = \sum_{n=1,3,\dots} \left\{ \frac{4V_s}{n\pi} \cos \frac{n\pi}{6} \right\} \sin n \left(\omega t + \frac{5\pi}{6} \right)$$

- For $n = 3$, $\cos \frac{n\pi}{6} = 0$. Hence, all the line voltages are free from triple 'n' ($3n$) Harmonics.

$$\begin{aligned} \text{RMS value of line voltage } (V_l) &= V_s \sqrt{\frac{120}{180}} \\ &= V_s \sqrt{\frac{2}{3}} \end{aligned}$$

$$V_l = 0.8165 V_s$$

- RMS value of phase voltage

$$(V_{ph}) = \frac{V_L}{\sqrt{3}} = \frac{0.8165 V_s}{\sqrt{3}} = 0.4714 V_s$$

120° Mode VSI:

$$V_{ao} = \sum_{n=1,3,\dots} \left\{ \frac{2V_s}{n\pi} \cos \frac{n\pi}{6} \right\} \sin n \left(\omega t + \frac{\pi}{6} \right)$$

$$V_{bo} = \sum_{n=1,3,\dots} \left\{ \frac{2V_s}{n\pi} \cos \frac{n\pi}{6} \right\} \sin n \left(\omega t - \frac{\pi}{6} \right)$$

$$V_{co} = \sum_{n=1,3,\dots} \left\{ \frac{2V_s}{n\pi} \cos \frac{n\pi}{6} \right\} \sin n \left(\omega t + \frac{5\pi}{6} \right)$$

- RMS Value of phase voltage

$$(V_{ph}) = \frac{V_s}{2} \sqrt{\frac{120}{180}} = 0.4082 V_s$$

$$V_{ph} = 0.4082 V_s$$

RMS value of line voltage

$$= V_{ph} (\sqrt{3}) = 0.4082 (\sqrt{3}) V_s = 0.7071 V_s$$

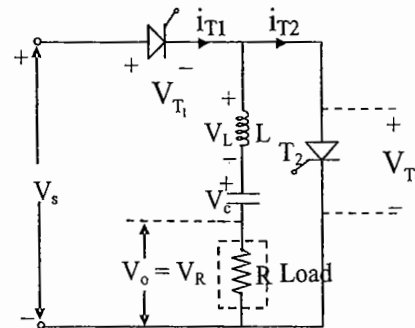
$$V_L = 0.7071 V_s$$

- For a given input dc supply, maximum output voltage can be obtained by using 1 - ϕ Full bridge even compared to 3 - ϕ inverters.

Series Inverter:

- Inverter in which commutating components are permanently connected in series with the load are called series inverters.
- The series circuit so formed must be under damped.

The Basic 1 - ϕ Series Inverter:



- The commutating components L and C values are so chosen such that the series RLC circuit forms an under damped circuit.
- When thyristor T_1 is turned ON, with T_2 off, current i starts building up in the RLC circuit; As the circuit is under damped, the load current after reaching some peak value, decays to zero at point a.
- SCR T_1 is turned off after instant a, some time t_{qmin} must be elapsed for T_1 to regain its forward blocking capability
- After T_1 has commutated, upper plate of capacitor attains positive polarity. Now when T_2 is turned on at instant b, capacitor begins to discharge and load current in the reversed direction builds up to some peak negative value and then decays to zero at instant c.
- Capacitor delivers the required energy during the negative half cycle.

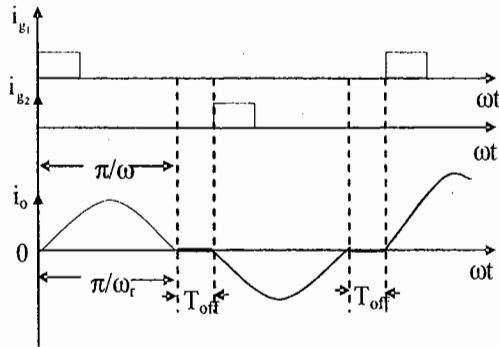
$$T_{off} > t_q \Rightarrow T_{off} = \left(\frac{\pi}{\omega} - \frac{\pi}{\omega_r} \right) > t_q$$

$$T_{off} = \frac{1}{2} \left(\frac{1}{f} - \frac{1}{f_r} \right)$$

Where, f = output frequency

f_r = Ringing frequency

T_{off} = ab or cd is called circuit turn-off time or dead- zone time.



Workedout Example

10. In a single-phase series inverter, the operating frequency is 50 kHz and the thyristor turn-off time $t_q = 10\mu s$. Circuit parameters are $R = 2\Omega$, $L = 60\mu H$, $C = 7.5\mu F$ and $V_s = 230V$ d.c, find circuit turn-off time

- (a) $15.32\mu sec$ (b) $17.83\mu sec$
(c) $19.23\mu sec$ (d) $21.46\mu sec$

Sol: operating frequency $\omega = 2\pi \times 5000$ rad/sec

$$\xi = \frac{R}{2L} = \frac{2 \times 10^6}{2 \times 60} = 0.0167 \times 10^6$$

$$\omega_0 = 1/\sqrt{LC} = \frac{106}{\sqrt{60 \times 7.5}} = 0.0471 \times 10^6$$

Circuit ringing frequency

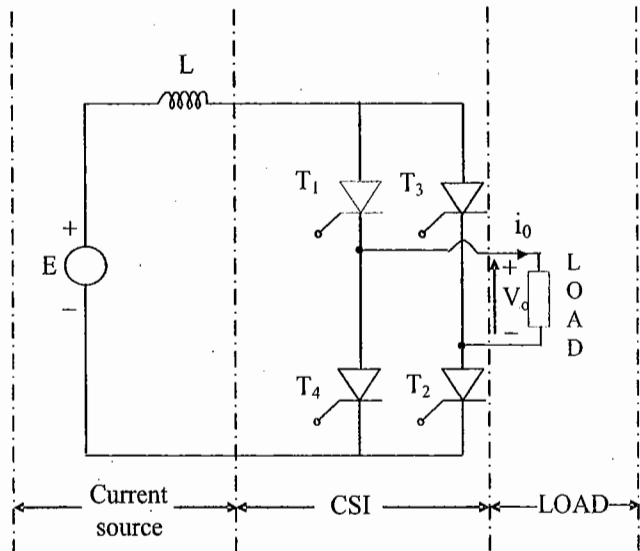
$$\omega_r = \sqrt{\omega_0^2 - \xi^2} = 0.044 \times 10^6 \text{ rad/sec}$$

Circuit turn-off time

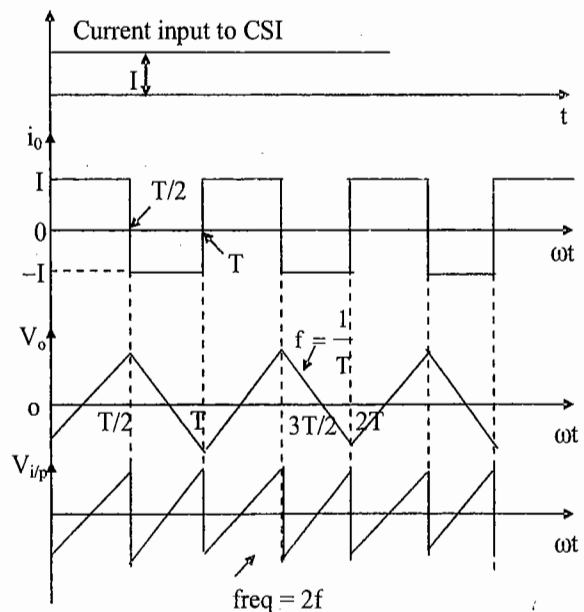
$$t_c = \frac{\pi}{\omega} - \frac{\pi}{\omega_r} = \frac{\pi}{2\pi \times 5000} - \frac{\pi}{0.04 \times 10^6}$$

$$= 21.46\mu sec$$

Current Source Inverter (CSI):



- During +ve half of current T_1, T_2 conducts.
During -ve half of current T_3, T_4 conducts.



- Current source inverter is suitable for capacitive loads. where as voltage source inverter is suitable for inductive loads.

$$V_p = \frac{I}{C} \times \left(\frac{T}{4} \right)$$

$$V_{p-p} = \frac{I}{C} \times \left(\frac{T}{2} \right)$$



Summary:

- If the current source inverter is feeding the capacitive load, then the following points are applicable:
- shape of the output voltage is triangular.
- shape of the output current is square wave.
- slope of the voltage variation is $\frac{I}{C}$.
- peak value of output voltage is $\left[\left(\frac{I}{C} \right) \left(\frac{T}{4} \right) \right]$
- peak to peak voltage is $\left[\left(\frac{I}{C} \right) \left(\frac{T}{2} \right) \right]$.
- If T_1, T_2 are conducting ; $V_o = V_{in}$
 T_3, T_4 are conducting; $V_o = -V_{in}$.
- In order to obtain output voltage frequency ' f ', the input voltage frequency should be ' $2f$ '.

Workedout Example

11. A single -phase auto-sequential commutated CSI is fed from 230V dc source, the load is $R = 20\Omega$. Thyristor have turn-off time of $20\mu\text{sec}$ and inverter output frequency is 50Hz. Take a factor of safety of 2.

Find source inductance and commutating capacitor assume maximum current change of 0.5A in one cycle.

- (a) 8.8H, $5.77\mu\text{F}$ (b) 9.2H, $2.88\mu\text{F}$
(c) 8.8H, $2.88\mu\text{F}$ (d) 9.2H, $5.77\mu\text{F}$

Sol: A short circuit at the load terminal of the inverter puts the most severe conditions on the source. So the value of source inductance must be obtained from this consideration.

$$V_s = L \frac{di}{dt}$$

$$\text{Where, } \frac{di}{dt} = \frac{0.5}{T} = 0.5 \times 50 = 25 \text{ A/sec}$$

$$\therefore L = 230/25 = 9.2 \text{ H}$$

$$\text{Circuit turn-off time } t_c = RC \ln 2$$

$$20 \times 2 \times 10^{-6} = 20 \times C \ln 2$$

$$C = \frac{2 \times 10^{-6}}{\ln 2} = 2.88 \mu\text{F}$$